INTERNAL TECHNICAL EVALUATION · RUN FULL-2026-10-05

Strands Decider vs TypeSafe Jev

A workload-fit, operational-readiness, and economics comparison for decision workloads in agent and business workflows.

Executive decision. Jev is the stronger general decision model and the safer default for tool gating, guardrails, and judging. Strands Decider is a credible open option for high-volume, plain intent routing, but the tested serving path is not production-ready: concurrent requests crash the server, and yes/no outputs require task-specific threshold calibration.

5 / 6
primary benchmark wins for Jev

3,080
Banking77 items where Decider won
84.1% vs 80.5%

$3.67
approximate study cost, internal only

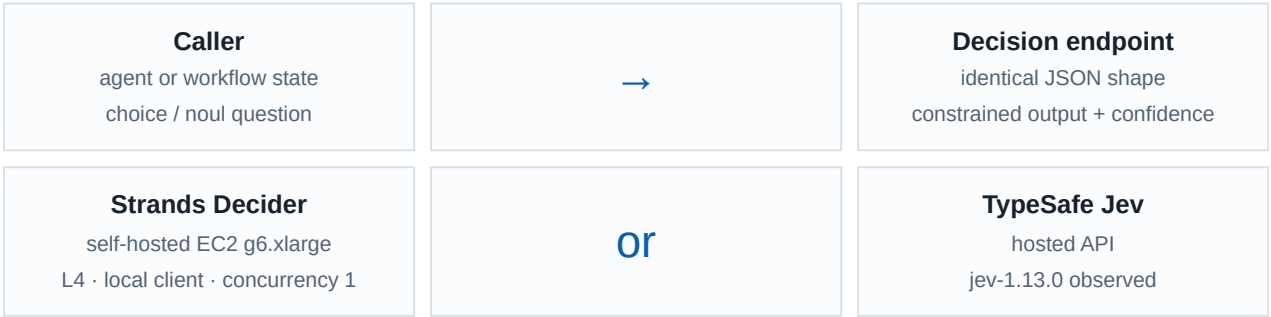
1. Decision summary

This study compares two systems that expose the same System One decision interface: a caller provides state and a constrained question, and the model chooses an option or returns a yes/no probability. The comparison is intentionally about classification, routing, tool-call gating, guardrails, and judging. It is not a test of open-ended generation or multi-step planning.

Workload	Measured result	Architecture implication
General decision quality	Jev wins JevBench public, 85.7% vs 72.7%, p=9e-6.	Use Jev as the general-purpose decision layer.
Tool-call gating	Jev 82.1% vs Decider 65.1%; tool hallucination 8.9% vs 26.2%.	Do not use the tested Decider path as an unguarded agent gate.
Plain intent routing	Decider wins Banking77, 84.1% vs 80.5%. On CLINC, Jev leads 89.5% vs 82.2%.	Decider is viable for selected high-volume routing workloads after hardening.
Binary guardrails and judging	Jev leads injection F1 0.681 vs 0.095 and HaluEval F1 0.864 vs 0.432.	Decider needs labelled calibration and a per-task threshold policy.
Operations	Concurrent Decider requests caused a sticky CUDA assert. Jev completed 17,116 calls with no errors.	Queue or fix concurrency before production adoption.
Input modality capability, not measured	Decider accepts images with `--vision`; Jev 1.13 is text-only. Jev accepts long states up to 32k tokens; Decider is capped at 4,096.	Choose by input type: image/screenshot decisions favour Decider; long-context decisions favour Jev.

> **Recommendation:** adopt a workload-specific policy. Prefer Jev for safety-sensitive and agent-control decisions. Consider Decider for internal, high-volume, latency-sensitive intent routing only after concurrency, long-input handling, observability, and calibration gates are satisfied.

2. Systems and evaluation boundary



System	Version / deployment	Test condition
Strands Decider	`strands-decider-2B-hobson-v19` on Qwen Qwen3.5-2B-Base; GitHub serving commit `75c9fd3`; `--strict-window`; 4,096-token window; flash-linear-attention 0.5.2.	AWS EC2 g6.xlarge, NVIDIA L4, us-east-1. Client ran on the instance to remove tunnel overhead. Accuracy run at concurrency 1.
TypeSafe Jev	`jev-latest`, returning `jev-1.13.0` throughout the run.	Hosted API. Accuracy run from laptop at concurrency 4. Input payloads were byte-identical except for model.

The optional Nova/Bedrock baseline was not included in the final comparison. Claude on Bedrock was unavailable in the test account because of an organizational SCP.

3. Benchmarks and method

ID	Dataset / task	n	Primary question
B1	JevBench public, 18 families and three tiers	231	General System One decisions, including hard reasoning cases
B2	NVIDIA When2Call	3,652	Call a tool, ask for information, cannot answer, or answer directly
B3	Banking77	3,080	Customer-support intent routing across 77 classes
B4	CLINC150 plus out-of-scope	5,500	Intent routing and recognition of none-of-the-listed intents
B5	deepset prompt-injections	116	Prompt-injection guardrail classification
B6	HaluEval QA subset	1,000	Groundedness / hallucination judging

Protocol controls

- Prompts were frozen and pre-registered before the full run. Both models received the same decision payloads.
- Accuracy, macro-F1, calibration, selective accuracy at confidence thresholds, AUROC, latency, cost, and exact paired McNemar comparisons were recorded.
- Decider received warm-up requests. Errors after six retries were counted and excluded from accuracy. Ten When2Call inputs above 4,096 tokens were rejected cleanly by strict-window mode.
- End-to-end latency is directional only because Decider was measured on localhost and Jev over the internet from India. Decider server time is the fair model-only comparison.

> **Interpretation boundary:** this is one run per system. It supports workload-fit decisions, not universal model rankings. Jev training data is unknown. HaluEval is derived from HotpotQA, which is in the Decider training mix, and Decider was iterated against the public JevBench set.

4. Primary results

Benchmark	n	Decider	Jev	Paired result
JevBench public	231	72.7%	**85.7%**	Jev, p=9e-6
When2Call neutral	3,652	65.1%	**82.1%**	Jev, p=8e-90
When2Call labelled	3,652	64.1%	**83.0%**	Jev, p=5.7e-105
Banking77	3,080	**84.1%**	80.5%	Decider, p=4.1e-8
CLINC150	5,500	82.2%	**89.5%**	Jev, p=1.9e-46
Prompt injection	116	50.9%	**75.0%**	Jev, p=5.8e-8
HaluEval contamination risk	1,000	63.7%	**87.1%**	Jev, p=4.8e-44

JevBench profile

Tier	n	Decider	Jev
Easy	48	100%	100%
Original	72	90%	99%
Hard	111	50%	71%

The gap is concentrated in harder families such as multi-hop, long-policy, probability, and adversarial tasks. Both systems perform well on straightforward extraction, intent, and tool-selection subsets.

5. Workload-specific findings

5.1 Tool-call gating is the main risk

On When2Call, Decider correctly identified tool calls 82.6% of the time and requests for missing information 78.4%, but recognised “cannot answer with the available tools” only 36.9% of the time. When no tools were offered, it was correct on 46.1% of items. It selected a tool when it should not have on 26.2% of items, compared with Jev at 8.9%.

> **Control recommendation:** if Decider is used in an agent loop, add an explicit policy layer that validates tool eligibility, required arguments, tool availability, and an abstain/cannot-answer path. Do not treat the model's top option as authorization by itself.

5.2 Intent routing is the strongest Decider fit

Decider beat Jev on Banking77 and had lower ECE, 0.032 versus 0.091. On CLINC it trailed overall and was weaker at out-of-scope recall, 0.612 versus 0.852, although its out-of-scope precision was higher, 0.944 versus 0.873. This suggests a conservative router that can work well when the class inventory is stable and abstention is acceptable.

5.3 Binary outputs require calibration

At the default 0.5 threshold, Decider's positive F1 was 0.095 on injection and 0.432 on HaluEval. Its AUROC was materially better, 0.870 and 0.911, showing ranking signal rather than reliable default thresholding. The post-hoc best thresholds were fitted on the test set and are optimistic. A production implementation needs a held-out calibration set per question type.

6. Modalities and input limits

The two systems are not equivalent on input. This is a capability boundary that should drive architecture as much as the accuracy tables do. It was **not** benchmarked in this study; the figures below come from vendor and project documentation.

Dimension	Strands Decider v19	Jev 1.13
Input modality	Text; images when served with `--vision` (no weight change; uses the vision tower in the base checkpoint).	Text only. Non-text inputs must be pre-processed into text or structured fields before being sent as `state`.
Image handling	Up to 4 images per request, each scaled down so the longer side is at most 448 px (~196 tokens for a square image). Images count against the same 4,096-token window. A text-only server rejects image requests with HTTP 422.	None. Images, audio, video, and binaries are unsupported and must be converted upstream.
Context window	4,096 tokens covering state, images, and all questions.	64k tokens per request; 32k for `state` plus the longest question. Multiple questions share one ingested state.
Weights and hosting	Open checkpoint, self-hosted; vision tower adds about 331M parameters (~0.66 GB in bf16).	Hosted only, same weights for every account, shaped through the request rather than per-account adaptation.

> **Unverified capability claims.** The Decider project reports vision accuracy of 0.784 on NaturalBench, 0.877 on POPE-adversarial, and 37/60 on the Image-JevBench preview (one run, on an H100; not reproduced here). The same document notes that with the image removed the model still answers at a mean confidence of 0.65–0.73, so its image-conditioned confidence is not a strong grounding signal on its own. Treat these numbers as vendor-reported until independently reproduced.

> **Architecture implication:** for screenshot or photo triage, Decider is the only one of the two that can take the image directly. For long policy documents, transcripts, or multi-document states, Jev's 64k window is the deciding factor. Where an image decision is needed and the model must be Jev, a separate vision-to-text step has to be designed, owned, and monitored.

7. Confidence, latency, and economics

Measure	Decider	Jev	Decision meaning
Banking77 p50 / p95	154 / 157 ms	305 / 478 ms p50 / p99	Decider is lower latency in this topology.
Model server time	66–237 ms p50 by task	Not exposed	Only Decider exposes model-side timing.
Achieved throughput	4.2–6.5 decisions/s, concurrency 1	~12–13 decisions/s, concurrency 4	Concurrency crash dominates Decider's operational profile.
Cost per 1,000	$0.015–$0.053 at full utilisation	$0.015–$0.106 measured input-token basis	Unit economics depend on utilisation and prompt size.
Fixed infrastructure	$0.8048/hour, about $588/month always-on	No GPU fixed cost	Self-hosting requires sustained load to amortise the GPU.

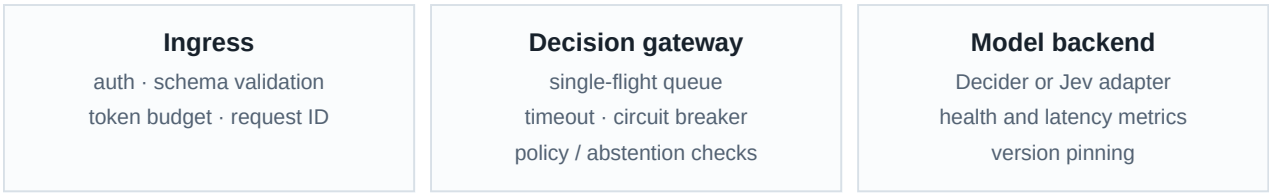
At the measured prompt sizes, the estimated break-even is approximately 6–10 million Jev decisions per month. One Decider box can serve at most about 13 million per month sequentially, and that comparison assumes full utilisation. This makes self-hosting economically attractive only for steady, high-volume workloads and only after serving reliability is fixed.

Confidence is useful but conservative

On Banking77, Decider was 97.8% accurate at confidence ≥0.9, but coverage was 48.3%. On When2Call, it was 97.4% accurate at that threshold with only 4.2% coverage. Jev had much higher coverage but was below 95% accuracy at ≥0.9 on several tasks. Confidence should therefore drive selective routing, not unconditional trust.

8. Operational readiness and target architecture

Finding	Impact	Required gate
Concurrent requests caused sticky CUDA device-side assert.	Service becomes unhealthy under parallel load.	Fix upstream or place a tested single-flight queue in front; add soak and failure-recovery tests.
PyPI 0.1.0 long prompts could crash the GPU.	One malformed/long request could poison later requests.	Enforce request token limits before inference; retain strict-window behavior and return a typed rejection.
Cold start was approximately 1–12 minutes after instance start.	Unsuitable for latency-sensitive scale-to-zero without warming.	Keep a warm capacity policy or accept startup SLO explicitly.
Causal-conv1d used a fallback kernel.	Measured latency is not a best-case optimized path.	Re-measure after compatible kernel/toolchain support.
Binary threshold is task-dependent.	Default yes/no policy creates false negatives.	Held-out calibration, threshold registry, drift monitoring, and rollback.



For a Decider deployment, the minimum production shape is not a direct caller-to-GPU connection. It is a decision gateway that validates inputs, enforces the context window, serializes or safely schedules requests, checks the selected action against policy, and exposes health, error, confidence, and calibration telemetry.

9. Recommendation and next steps

Recommended now

1. **Use Jev** for production agent gating, guardrails, groundedness judging, and general decision workloads.
2. **Evaluate Decider selectively** for stable, high-volume intent routing where self-hosting, data locality, or predictable model-side latency is a requirement.
3. **Do not claim cost savings from self-hosting by default.** The GPU only wins economically near sustained high utilisation.

Promotion gates for Decider

Gate	Acceptance criterion
Reliability	Parallel load, queue saturation, restart, and malformed/over-window inputs complete without sticky GPU failure.
Quality	Meet workload-specific accuracy and false-action thresholds on a held-out representative set, not only public datasets.
Calibration	Per-task thresholds selected on validation data; confidence coverage and error rates monitored in production.
Economics	Forecasted utilisation exceeds the self-hosting break-even after capacity, storage, operations, and redundancy costs.
Governance	Versioned model/checkpoint, dataset lineage, audit logs, access controls, and rollback procedure are documented.

Limitations

- One run per backend. No retraining or seed variance estimate.
- End-to-end latency paths are not equivalent.
- HaluEval contamination and JevBench selection pressure limit generalisation claims.
- Prompt-injection dataset is small and uses a broad definition that includes role-play prompts.
- Vision capability and accuracy are vendor-reported and were not reproduced in this study; only text decisions were benchmarked.
- Tic-tac-toe pilot showed neither model plans reliably; these are decision classifiers, not game planners.